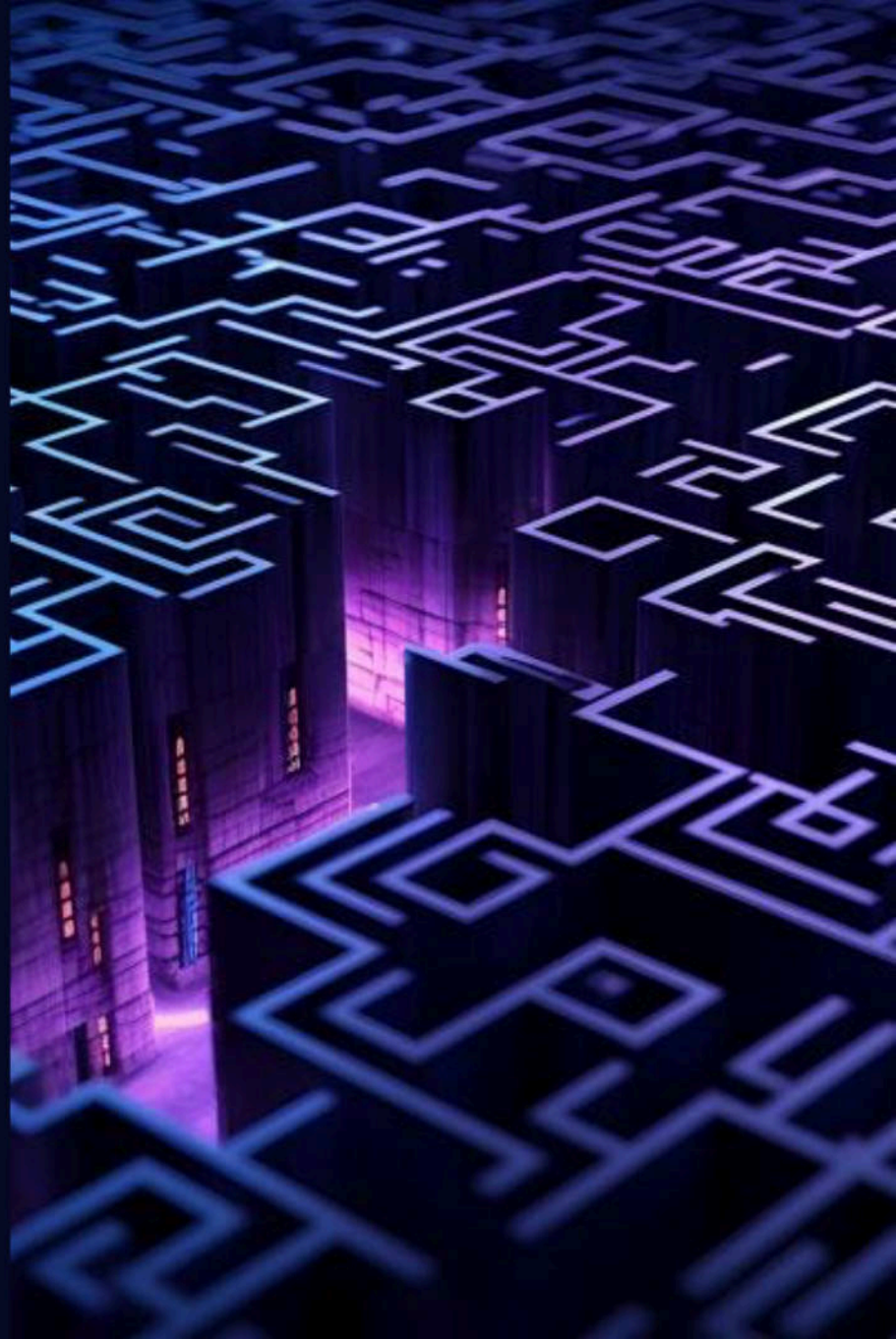
Lexfo

ICONV, set the charset to RCE

EXPLOITING THE GLIBC TO HACK THE PHP ENGINE

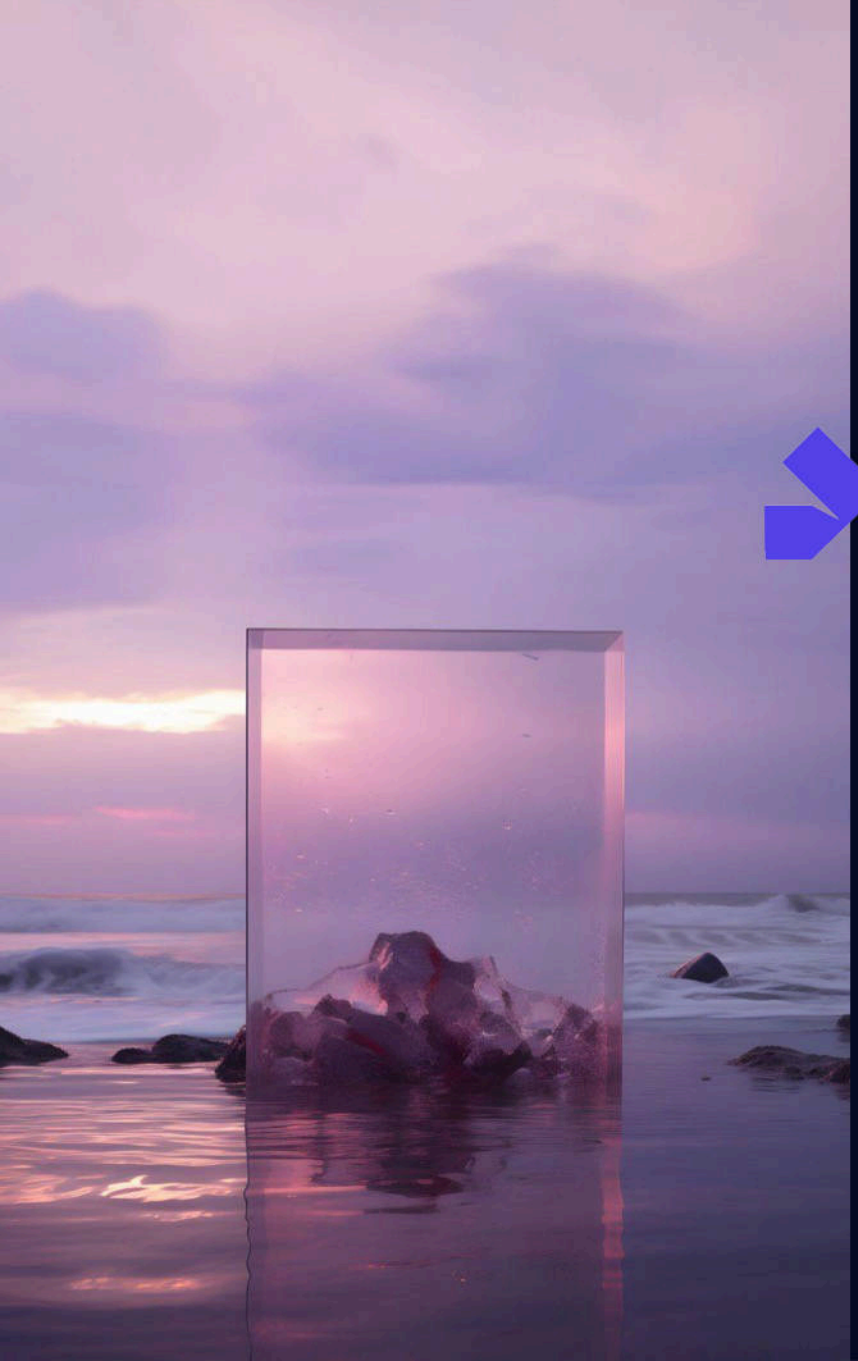
Charles Fol
@cfreal_



DEFCON 32 ENGAGE

Summary

Lexfo



Discovery

Lexfo

PHP file read primitive

What can you do with a file read primitive?

```php
echo file_get_contents($_GET['file']);
```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```php
echo file_get_contents('http://ambionics.io/blog');
```


PHP file read primitive

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```

```php
echo file_get_contents('/etc/passwd');
```

```php
echo file_get_contents('http://ambionics.io/blog');
```

```php
echo file_get_contents('ftp://ftp.ambionics.io/file.bin');
```


PHAR : a dangerous protocol

```
phar:///var/www/archive.phar/file.txt
```

PHAR: PHP archive

- Source files
- Resources
- Serialized metadata
  - Deserialized when reading an archive

PHAR : File read to RCE

Harder to exploit

- PHAR:
  - PHP 8+: no more deserialisation of the metadata
  - phar:// might be disabled (Drupal, Magento)

- Deserialisation:
  - Gadget chains are being patched more and more
  - Types are making a comeback

PHP filters: applying transforms to the stream

```
php://filter/[filters]/resource=[resource]
```

`[resource]`	original stream: file, etc.
`[filters]`	list of filters to apply to the stream

PHP filters: applying transforms to the stream

```
php://filter/convert.base64-encode/resource=/etc/passwd
```

PHP filters: applying transforms to the stream

```
php://filter/convert.base64-encode|convert.base64-encode/resource= ...
```

PHP filters: lots of options

Many available filters

- string.upper *Convert stream to uppercase*
- string.lower *Convert stream to lowercase*
- string.rot13 *Do some BC crypto*
- dechunk *Decode chunked encoding*
- convert.iconv.X.Y *Convert from charset X to charset Y*

PHP filters: applying transforms to the stream

```
php://filter/convert.iconv.UTF-8.UTF-16/resource=/etc/passwd
```

```
00000000: fffe 7200 6f00 6f00 7400 3a00 7800 3a00  ..r.o.o.t.:.x.:.
00000010: 3000 3a00 3000 3a00 7200 6f00 6f00 7400  0.:.0.:.r.o.o.t.
00000020: 3a00 2f00 7200 6f00 6f00 7400 3a00 2f00  :./.r.o.o.t.:./.
00000030: 6200 6900 6e00 2f00 6100 7300 6800 0a00  b.i.n./.a.s.h...
00000040: 6200 6900 6e00 3a00 7800 3a00 3100 3a00  b.i.n.:.x.:.1.:.
00000050: 3100 3a00 6200 6900 6e00 3a00 2f00 6200  1.:.b.i.n.:./.b.
00000060: 6900 6e00 3a00 2f00 7300 6200 6900 6e00  i.n.:./.s.b.i.n.
00000070: 2f00 6e00 6f00 6c00 6f00 6700 6900 6e00  /.n.o.l.o.g.i.n.
00000080: 0a00 6400 6100 6500 6d00 6f00 6e00 3a00  ..d.a.e.m.o.n.:.
00000090: 7800 3a00 3200 3a00 3200 3a00 6400 6100  x.:.2.:.2.:.d.a.
000000a0: 6500 6d00 6f00 6e00 3a00 2f00 7300 6200  e.m.o.n.:./.s.b.
000000b0: 6900 6e00 3a00 2f00 7300 6200 6900 6e00  i.n.:./.s.b.i.n.
000000c0: 2f00 6e00 6f00 6c00 6f00 6700 6900 6e00  /.n.o.l.o.g.i.n.
000000d0: 0a00 6100 6400 6d00 3a00 7800 3a00 3300  ..a.d.m.:.x.:.3.
000000e0: 3a00 3400 3a00 6100 6400 6d00 3a00 2f00  :.4.:.a.d.m.:./.
000000f0: 7600 6100 7200 2f00 6100 6400 6d00 3a00  v.a.r./.a.d.m.:.
00000100: 2f00 7300 6200 6900 6e00 2f00 6e00 6f00  /.s.b.i.n./.n.o.
00000110: 6c00 6f00 6700 6900 6e00 0a00 6c00 7000  l.o.g.i.n...l.p.
00000120: 3a00 7800 3a00 3400 3a00 3700 3a00 6c00  :.x.:.4.:.7.:.l.
00000130: 7000 3a00 2f00 7600 6100 7200 2f00 7300  p.:./.v.a.r./.s.
00000140: 7000 6f00 6f00 6c00 2f00 6c00 7000 6400  p.o.o.l./.l.p.d.
```


PHP filters: arbitrary suffix

Arbitrary prefix and suffix? [1]

```php
$json = file_get_contents($_GET['file']);
$json = json_decode($json);

echo $json['message']['title'];
```

[1] https://www.ambionics.io/blog/wrapwrap-php-filters-suffix

PHP filters: finding the bug

Bruteforcing conversions

```
php://filter/convert.iconv.X.Y/resource=data:,test123
```

```
segmentation fault (core dumped)
```


Woops, a crash

iconv: glibc's character set conversion library

```
iconv_t iconv_open(const char *tocode, const char *fromcode);
size_t iconv(iconv_t cd,
             char **inbuf, size_t *inbytesleft,
             char **outbuf, size_t *outbytesleft);
```

Processes at most **inbytesleft** bytes from **inbuf** and writes at most **outbytesleft** bytes to **outbuf***

* in theory

The vulnerability: OOB write

glibc-2.39/iconvdata/iso-2022-cn-ext.c

```c
if ((used & SO_mask) != 0 && (ann & SO_ann) != (used << 8))
{
    if (outptr + 4 > outend)
    {
        result = __GCONV_FULL_OUTPUT;
        break;
    }

    assert(used >= 1 && used <= 4);
    escseq = ")A\0\0)G)E" + (used - 1) * 2;
    *outptr++ = ESC;
    *outptr++ = '$';
    *outptr++ = *escseq++;
    *outptr++ = *escseq++;

    ann = (ann & ~SO_ann) | (used << 8);
}
else if ((used & SS2_mask) != 0 && (ann & SS2_ann) != (used << 8))
{
    assert(used == CNS11643_2_set); /* XXX */
    escseq = "*H";
    *outptr++ = ESC;
    *outptr++ = '$';
    *outptr++ = *escseq++;
    *outptr++ = *escseq++;

    ann = (ann & ~SS2_ann) | (used << 8);
}
else if ((used & SS3_mask) != 0 && (ann & SS3_ann) != (used << 8))
{
    assert((used >> 5) >= 3 && (used >> 5) <= 7);
    escseq = "+I+J+K+L+M" + ((used >> 5) - 3) * 2;
    *outptr++ = ESC;
    *outptr++ = '$';
    *outptr++ = *escseq++;
    *outptr++ = *escseq++;

    ann = (ann & ~SS3_ann) | (used << 8);
}
```

When converting to ISO-2022-CN-EXT, the library fails to **check the bounds** before writing **escape sequences** to the output buffer

Overflow of 1 to 3 bytes:

```
$*H 24 2A 48
$+I 24 2B 49
$+J 24 2B 4A
$+K 24 2B 4B
$+L 24 2B 4C
$+M 24 2B 4D
```


The vulnerability: OOB write

glibc-2.39/iconvdata/iso-2022-cn-ext.c

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if ((used & SO_mask) != 0 && (ann & SO_ann) != (used << 8))
{
    if (outptr + 4 > outend)
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    }

    assert(used >= 1 && used <= 4);
    escseq = ")A\0\0)G)E" + (used - 1) * 2;
    *outptr++ = ESC;
    *outptr++ = '$';
    *outptr++ = *escseq++;
    *outptr++ = *escseq++;

    ann = (ann & ~SO_ann) | (used << 8);
}
else if ((used & SS2_mask) != 0 && (ann & SS2_ann) != (used << 8))
{
    assert(used == CNS11643_2_set); /* XXX */
    escseq = "*H";
    *outptr++ = ESC;
    *outptr++ = '$';
    *outptr++ = *escseq++;
    *outptr++ = *escseq++;

    ann = (ann & ~SS2_ann) | (used << 8);
}
else if ((used & SS3_mask) != 0 && (ann & SS3_ann) != (used << 8))
{
    assert((used >> 5) >= 3 && (used >> 5) <= 7);
    escseq = "+I+J+K+L+M" + ((used >> 5) - 3) * 2;
    *outptr++ = ESC;
    *outptr++ = '$';
    *outptr++ = *escseq++;
    *outptr++ = *escseq++;

    ann = (ann & ~SS3_ann) | (used << 8);
}
```

CVE-2024-2961

AAAAAA 劄

BBBBBB 砐

CCCCCC 岍

DDDDDD 湿

Tedious conditions

Vulnerability

- 1, 2 or 3 bytes overflow
- **fixed** byte values

Conditions

- Control the **output charset** (ISO-2022-CN-EXT)
- Control **part of the input buffer**
- Have a **suitable output buffer**

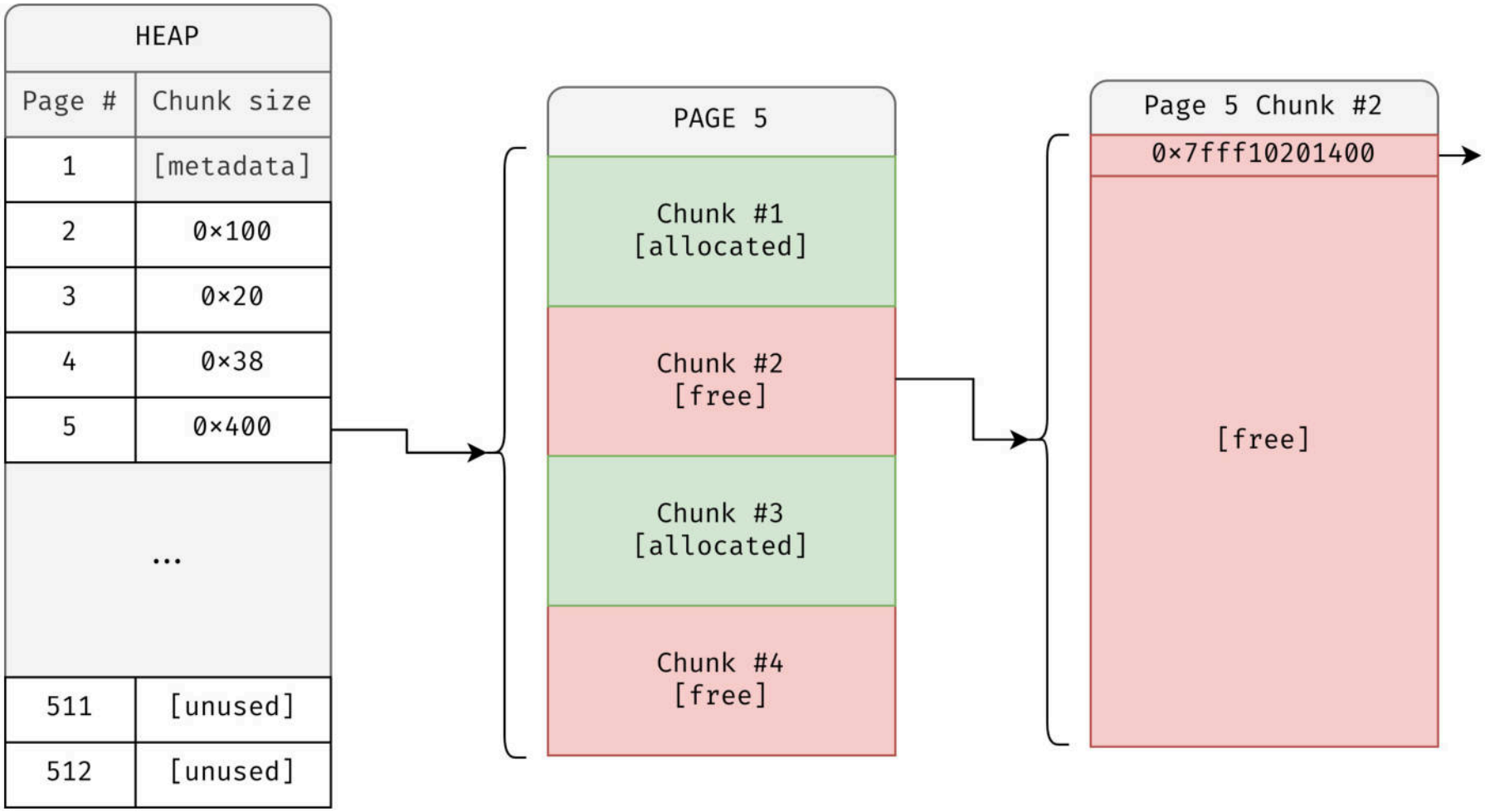
Attacking PHP filters

Lexfo

PHP's heap

- PHP's heap is page-based
  - 512 pages of 0x1000 bytes
- Each page contains chunks of the same size
  - Page 10: chunks of size 0x100
  - Page 11: chunks of size 0x30
  - Page 12: chunks of size 0xa00
- PHP holds a singly-linked free list for each chunk size
  - Last In, First Out (LIFO)
  - Similar to libc's tcache, but unlimited

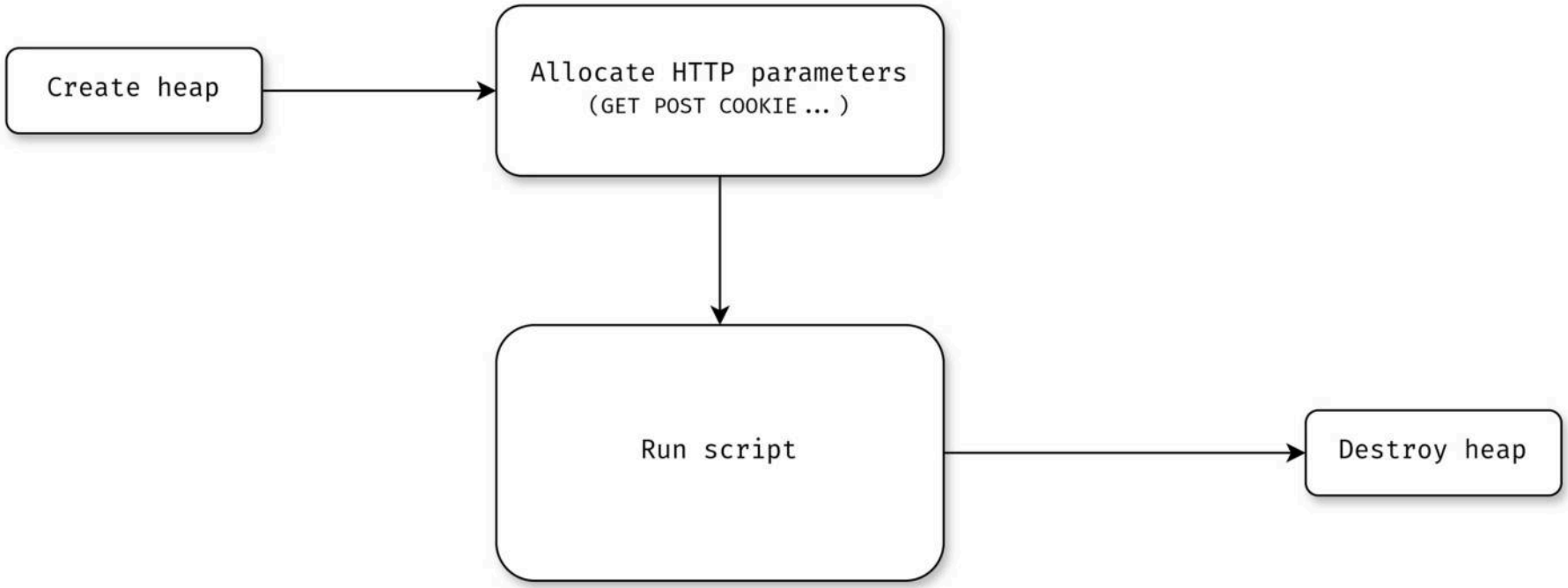
PHP's heap


HEAP
Page #
Chunk size
1
[metadata]
2
0×100
3
0×20
4
0×38
5
0×400
...
511
[unused]
512
[unused]
PAGE 5
Chunk #1
[allocated]
Chunk #2
[free]
Chunk #3
[allocated]
Chunk #4
[free]
Page 5 Chunk #2
0×7fff10201400
[free]

Request lifecycle

1 heap for 1 request

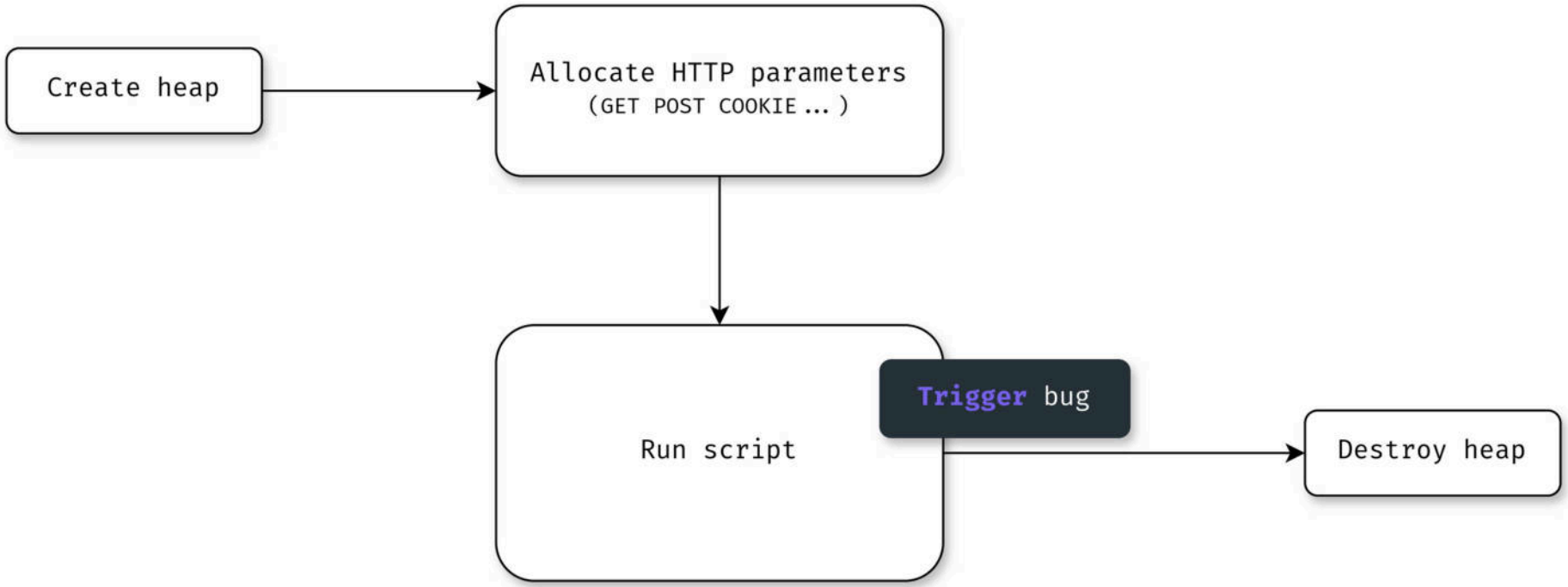
(bugs do not linger)



Request lifecycle

Triggering the **bug**

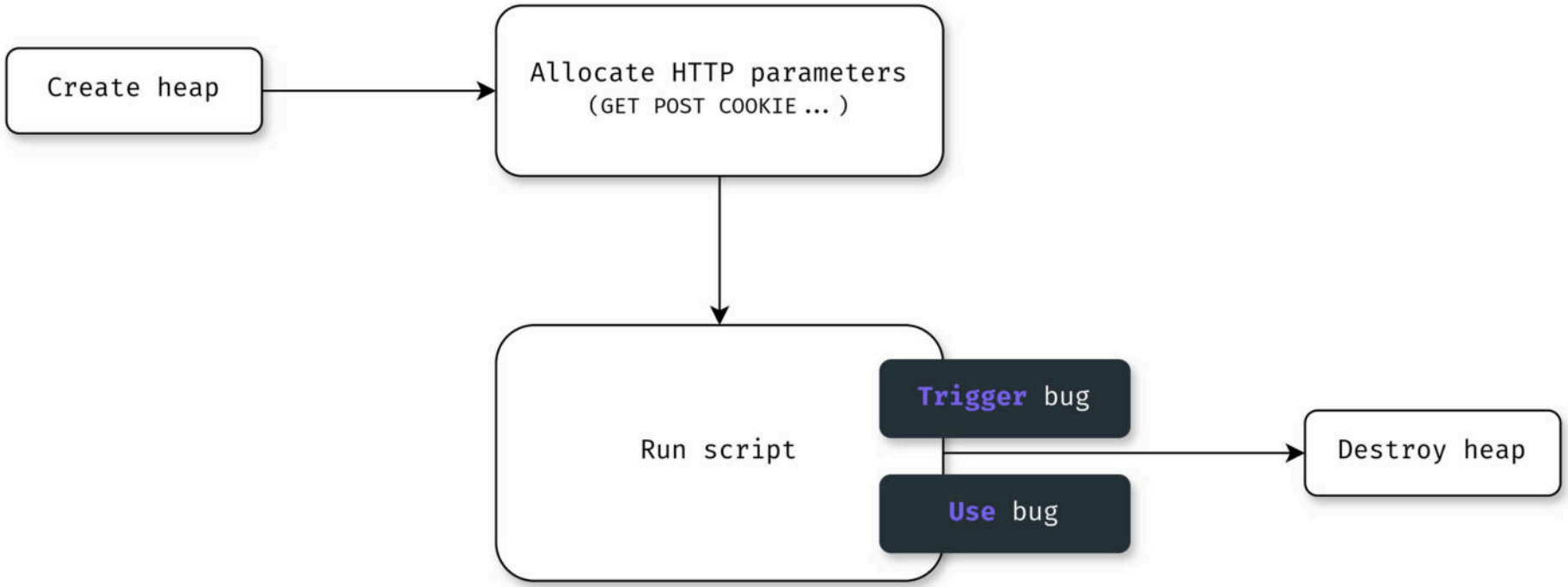
(finding a vulnerable code path)



Request lifecycle

Using the **bug**

(in the same request)



Handling of PHP filters

```
php://filter/[filters]/resource=[resource]
```

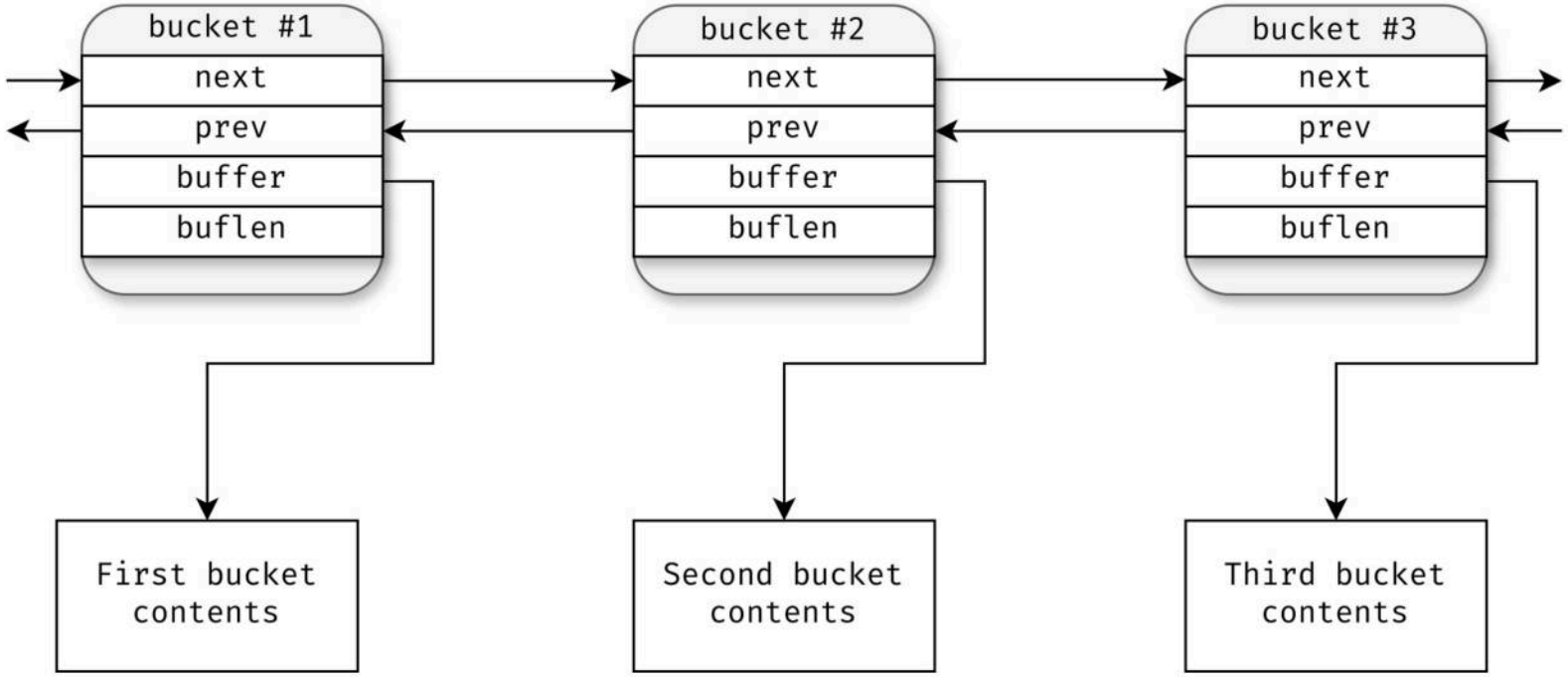
PHP processes a php://filter
in 2 steps

1. Read the resource as a stream
2. Apply filters

Handling of PHP filters

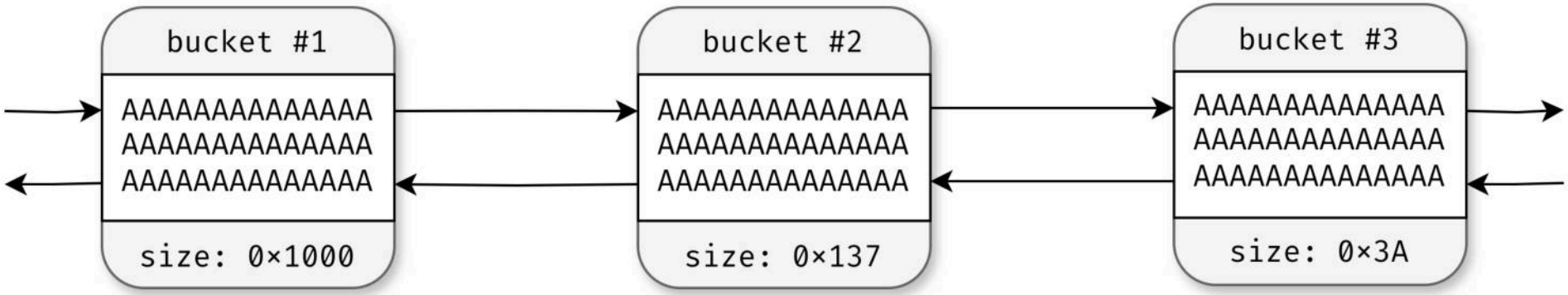
```
php://filter/[filters]/resource=[resource]
```

1. Fetching the stream



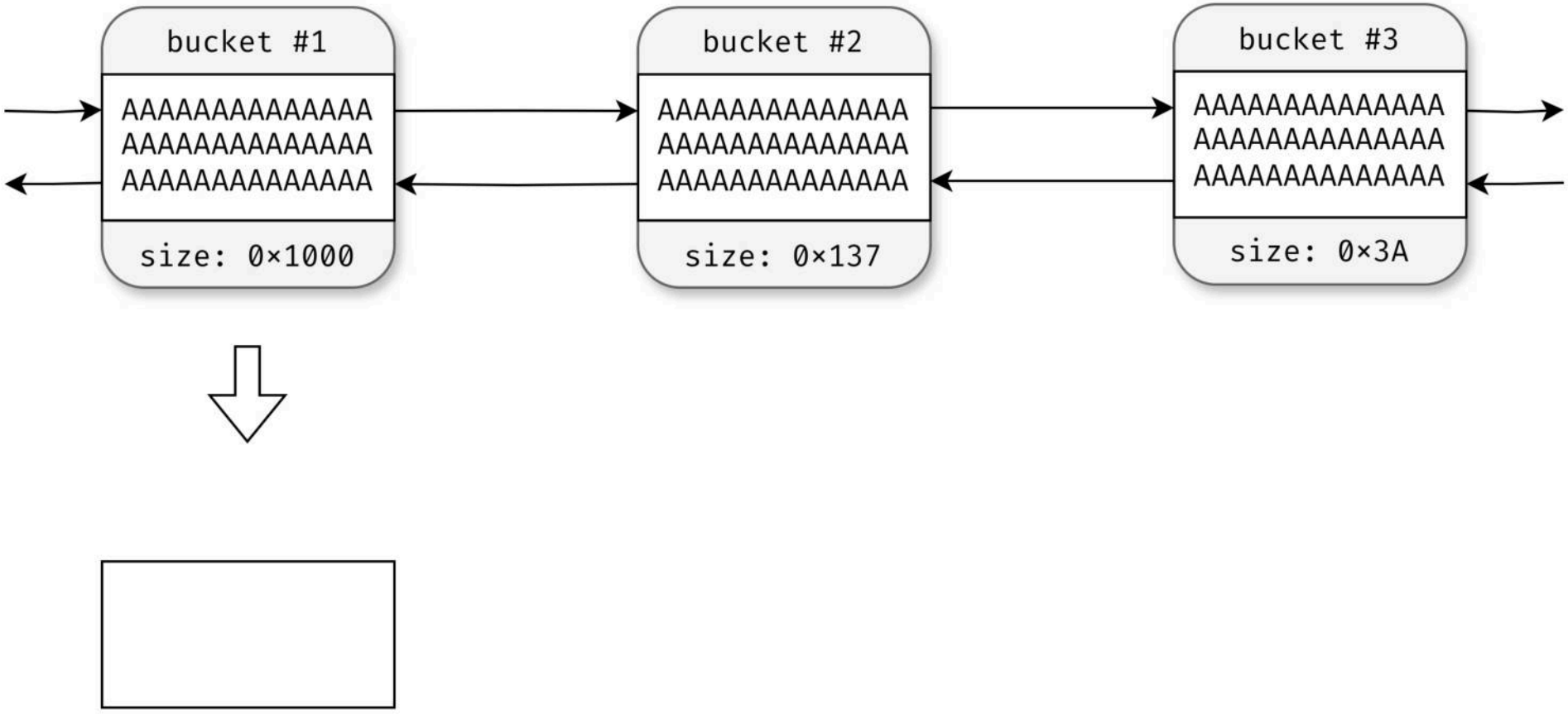
Handling of PHP filters

2. Applying filters



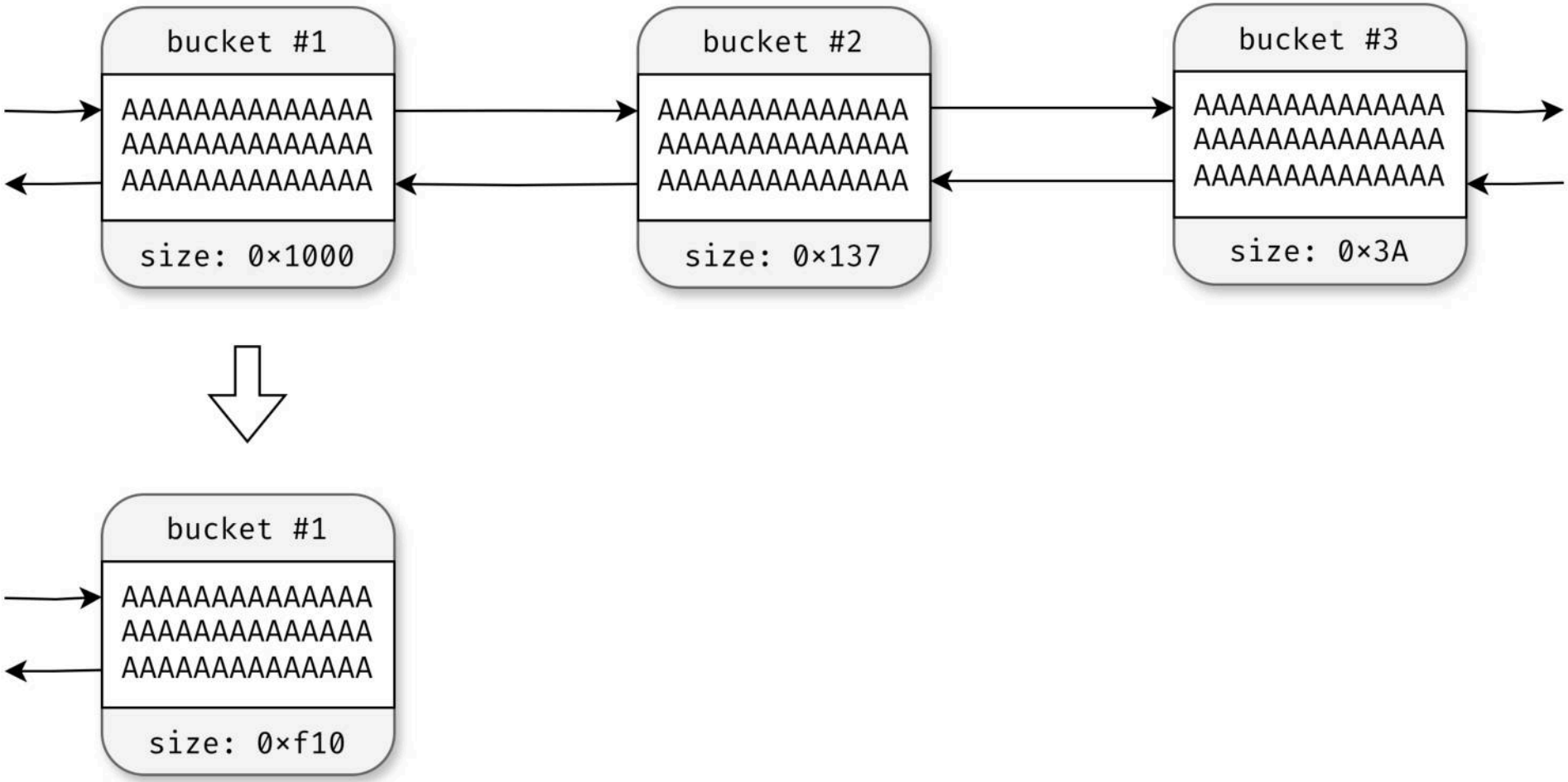
Handling of PHP filters

3. Applying filters

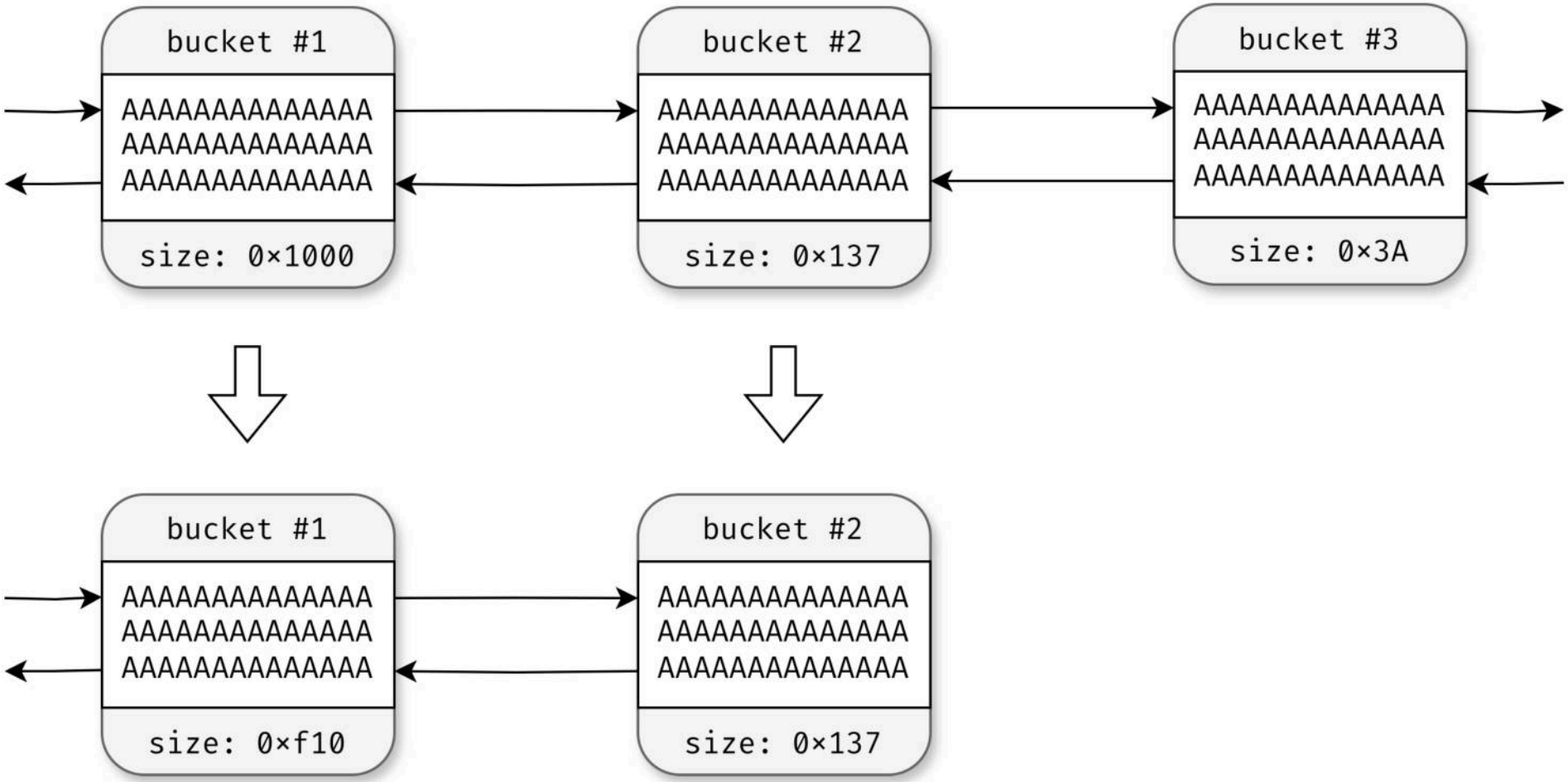


Handling of PHP filters

3. Applying filters



3. Applying filters



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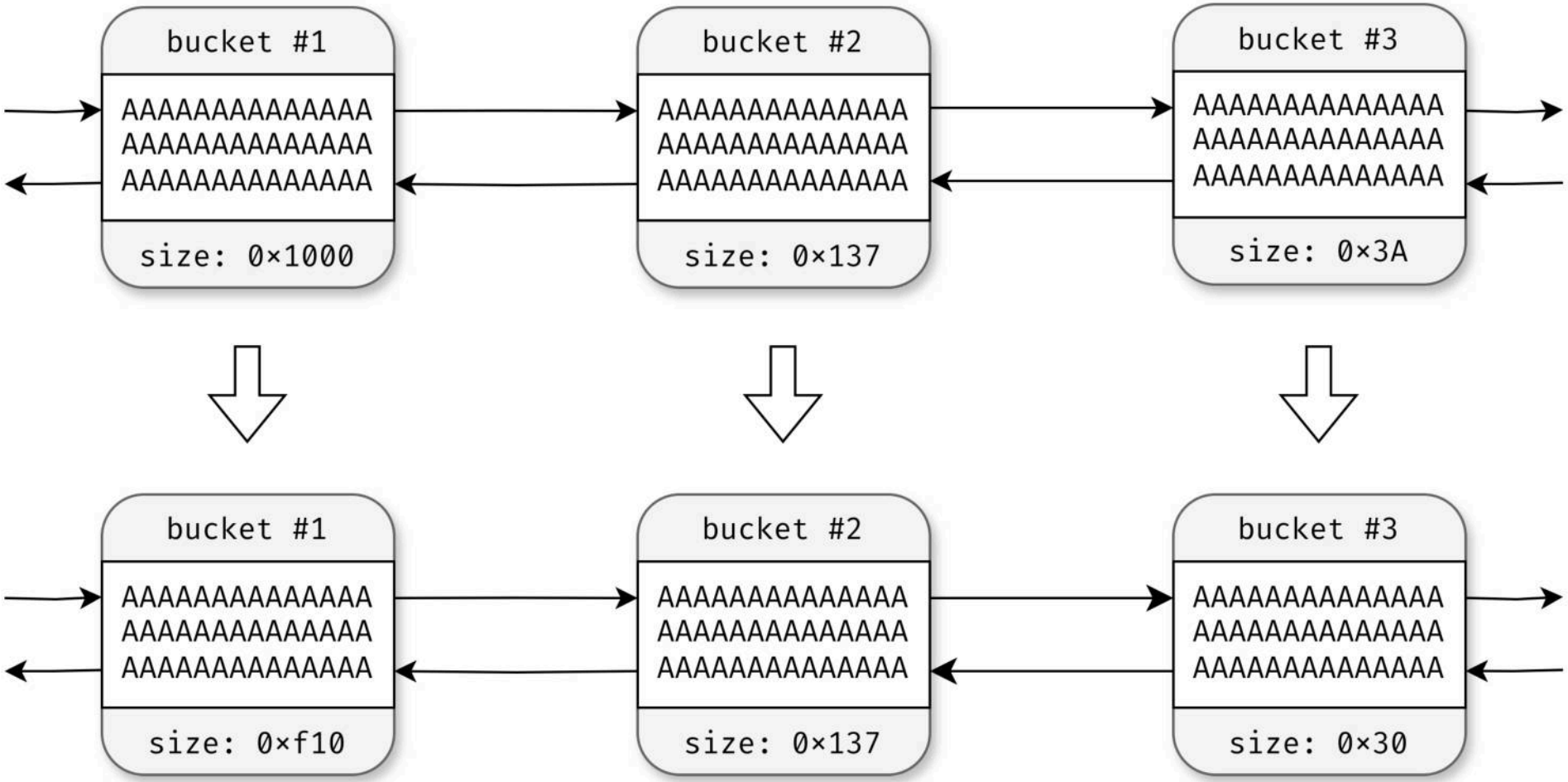


Building an exploit: file read

```php
echo file_get_contents($_GET['file']);
```

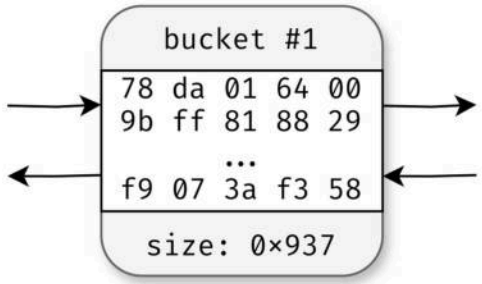
Not too hard ?

- Read binaries (php, libc)
- ASLR/PIE is NOT a problem
- Address of PHP's heap is known
- Arbitrary allocations/frees

But...

- Unknown environment:
  - Application:
    - Wordpress, Drupal, Symfony...
  - PHP version
  - Heap state
- Need a reliable exploit

Building an exploit: a single bucket

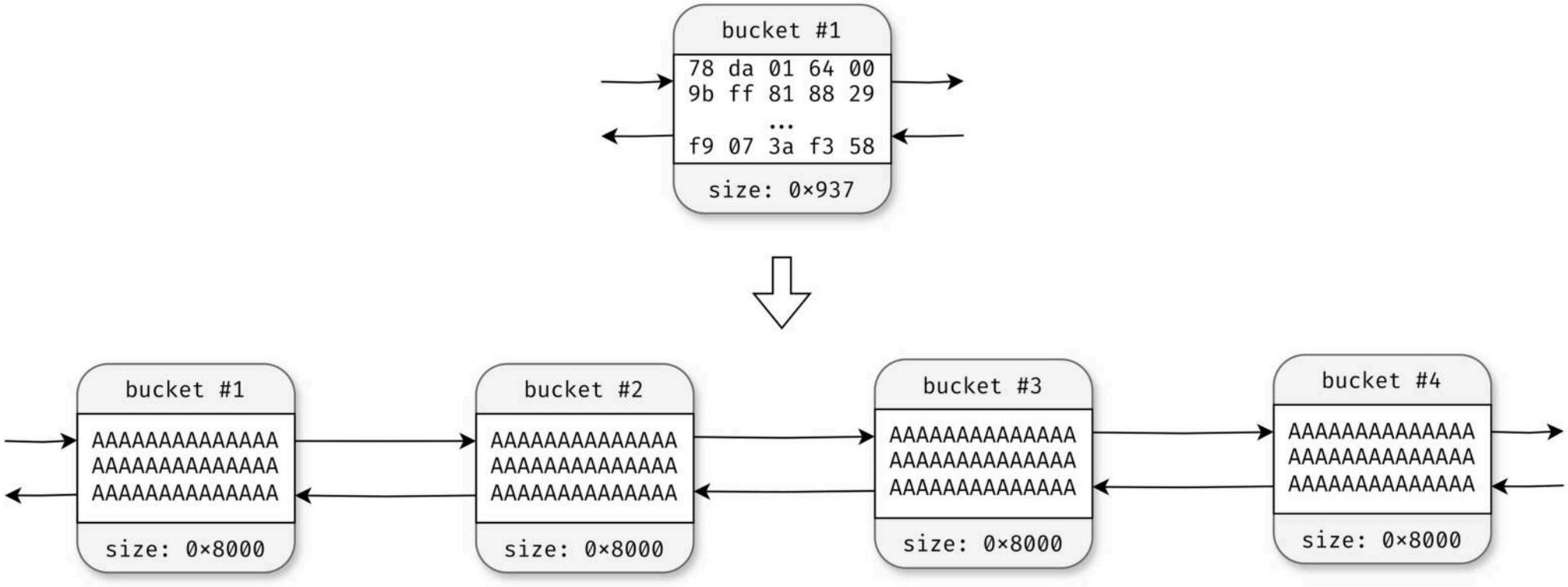


Streams are read in one bucket

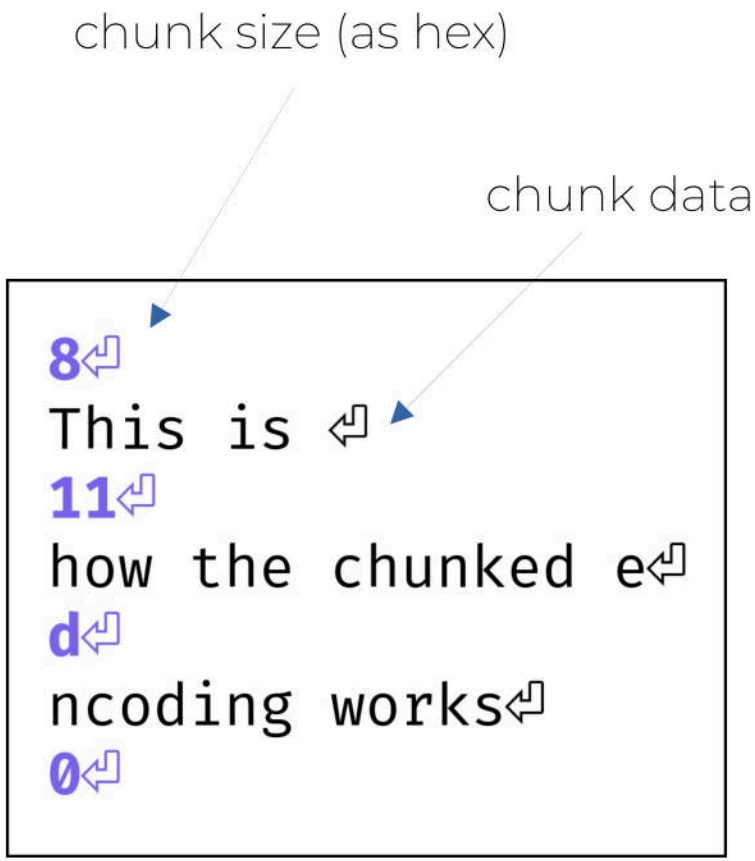
Impractical for exploitation

- Cannot :
  - Spray
  - Pad
  - Use altered free list

Building an exploit: using zlib.inflate to create buckets



Building an exploit: unfitting chunk size

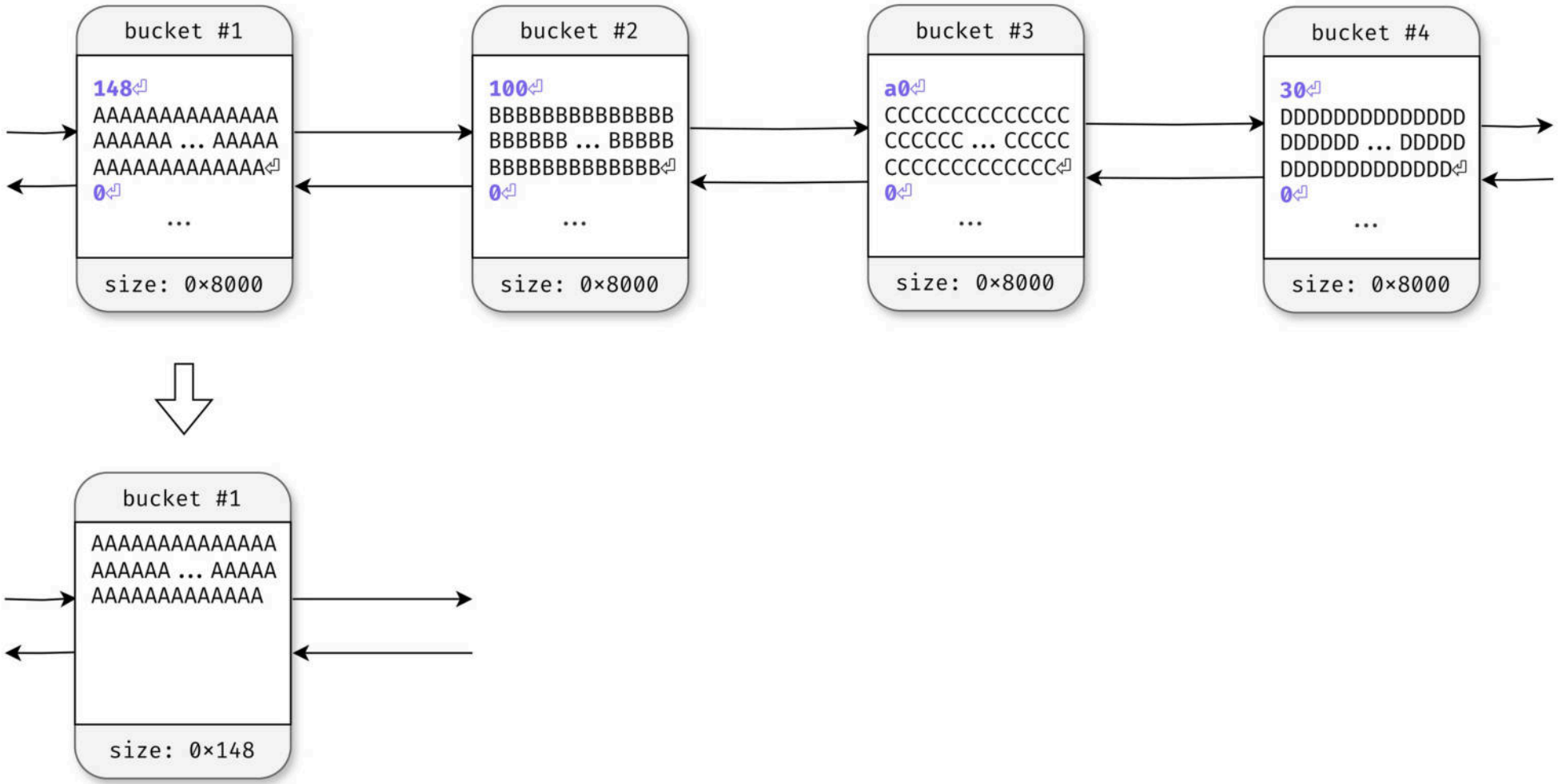


Resizing chunks to arbitrary sizes

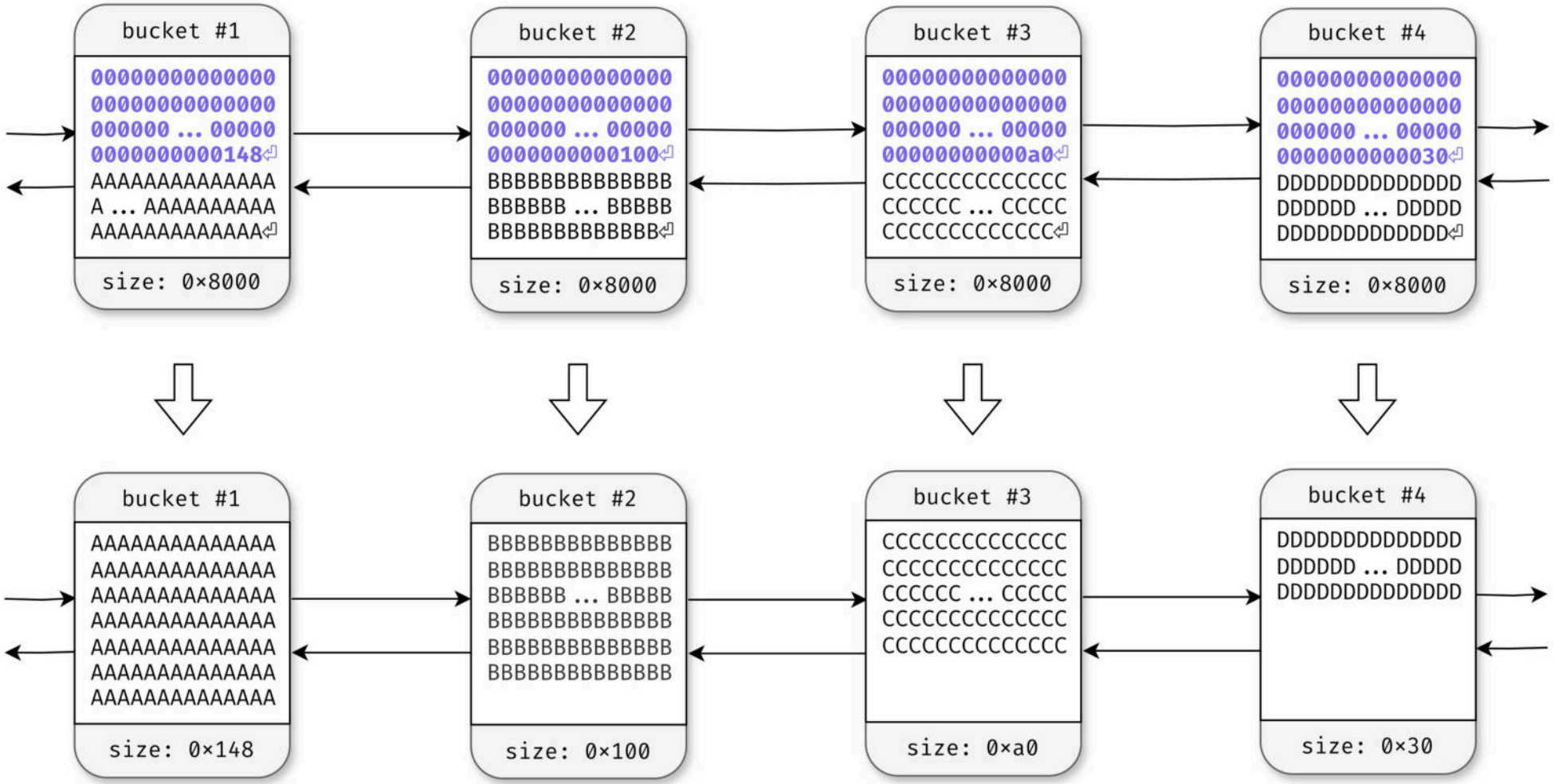
*Using the undocumented **dechunk** filter*

« Decodes a string which is HTTP-chunked encoded. »

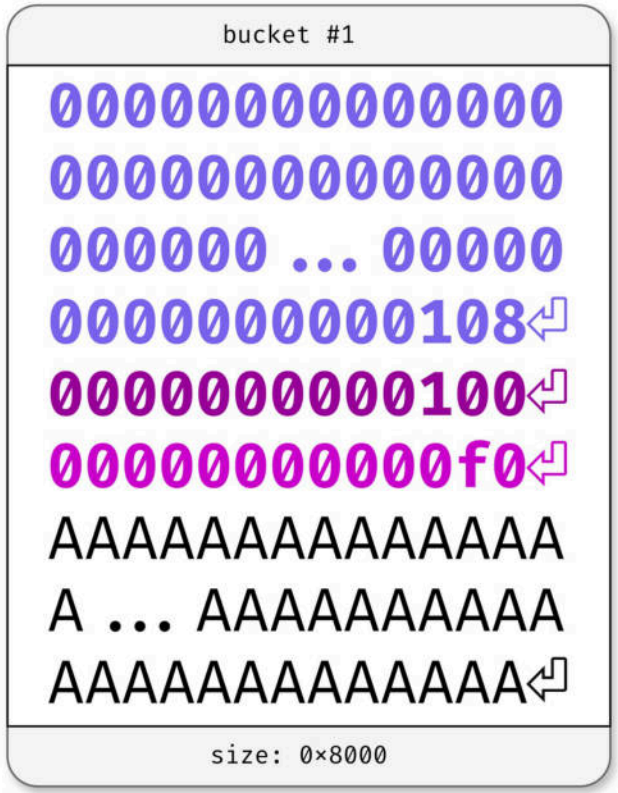
Building an exploit: unfitting chunk size (ox8ooo)



Building an exploit: using dechunk to change chunk sizes



Building an exploit: arbitrary allocations



Chunk attack

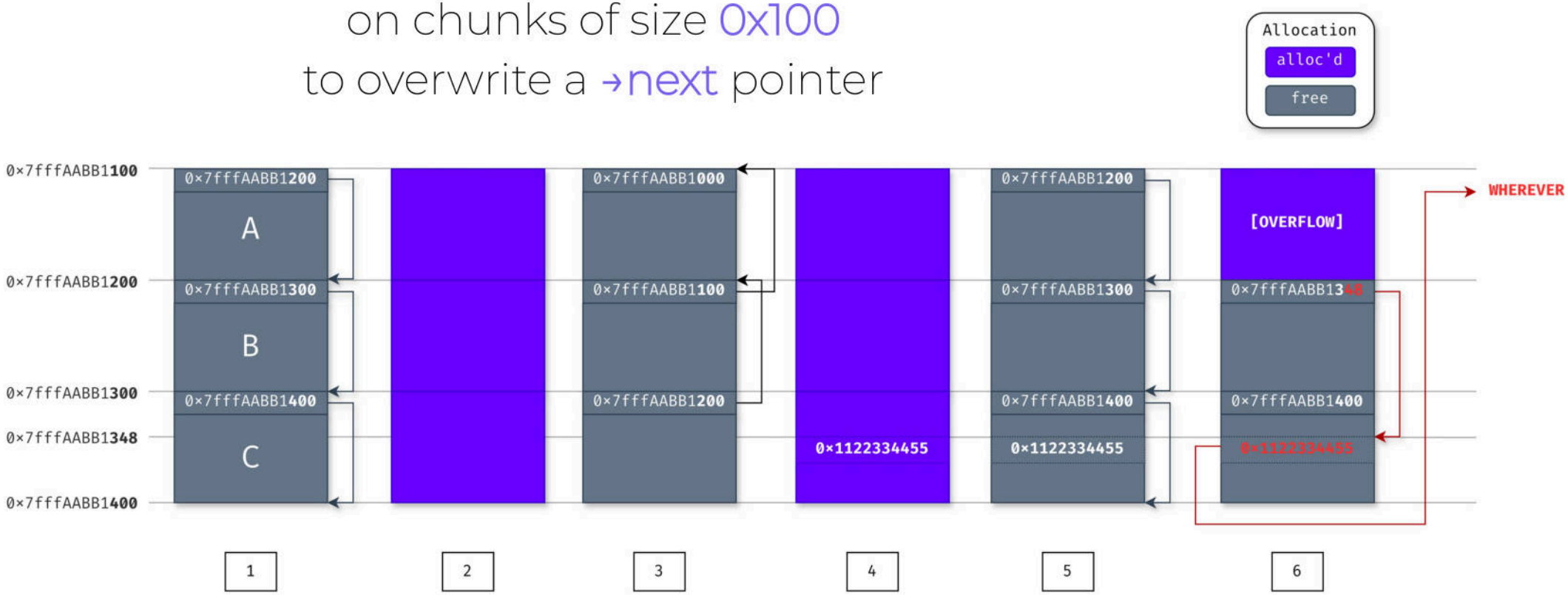
- We attack chunks of size 0x100
- Chunks of other sizes do **not** matter
- Because they are not in the same page

Dechunking buckets

- Use **dechunk** to make buckets appear/disappear
- Buckets appear when they are needed

Building an exploit: controlling a free list

1-byte overflow (48h)
on chunks of size 0x100
to overwrite a →next pointer



Building an exploit: endgame

php-8.3.2/Zend/zend_alloc.c

```
1   struct _zend_mm_heap {
2       // Whether to use custom allocation functions
3       int                use_custom_heap;
4
5       ...
6
7       // Free lists for each chunk size
8       zend_mm_free_slot *free_slot[ZEND_MM_BINS];
9
10      ...
11
12      // Custom allocation functions
13      struct {
14              void      *(*_malloc)(size_t);
15              void       (*_free)(void*);
16              void      *(*_realloc)(void*, size_t);
17      } custom_heap;
18
19      ...
20  };
```

Code execution

- 1-byte overflow (0x48) to alter a free list
- Allocate at arbitrary address
- Overwrite zend_mm_heap
  - At the top of the heap region
- Overwrite function pointers
  - free == system
- Call free(command)

Building an exploit

CNEXT performance

- ✔ Works against any target
  - ➔ From PHP 7+ to current (2015 to 2024)
  - ➔ Any PHP application
- ✔ 100% reliable
  - ➔ No crash
- ✔ Payload < 1000 bytes
  - ➔ Suitable for GET parameters
- ✔ Self-contained exploit
  - ➔ No additional HTTP parameters

File read → RCE

Demo: Magento XXE to RCE

CosmicSting

Magento XXE

(CVE-2024-34102)

June, 2024

IMPACT

Without cnext

Read files

Create API keys

With cnext

Remote

Code

Execution

https://sansec.io/research/cosmicsting

Exploiting direct iconv() calls

Lexfo

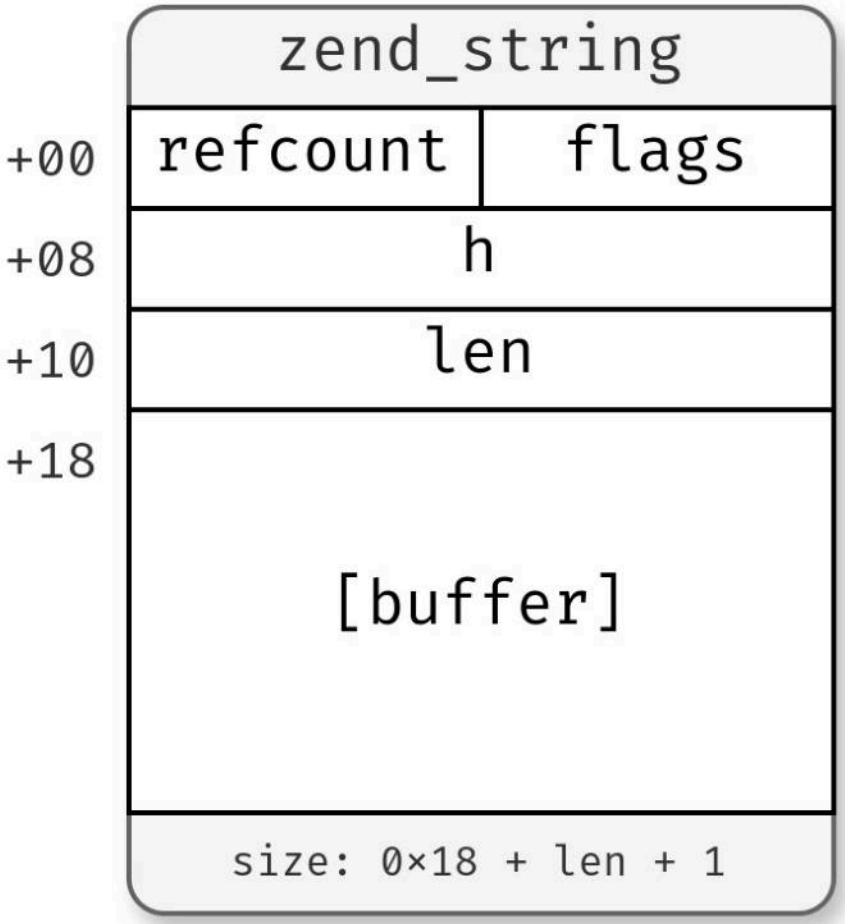
Another sink: iconv() calls

```
iconv(string $from_encoding, string $to_encoding, string $string): string

    iconv_substr( ... ) iconv_mime_decode( ... )
iconv_mime_decode_headers( ... ) iconv_strlen( ... )
    iconv_strpos( ... ) iconv_strrpos( ... )
```

- Direct calls to iconv*() are vulnerable
- Exploitation is completely different

Exploit strategy: getting a leak



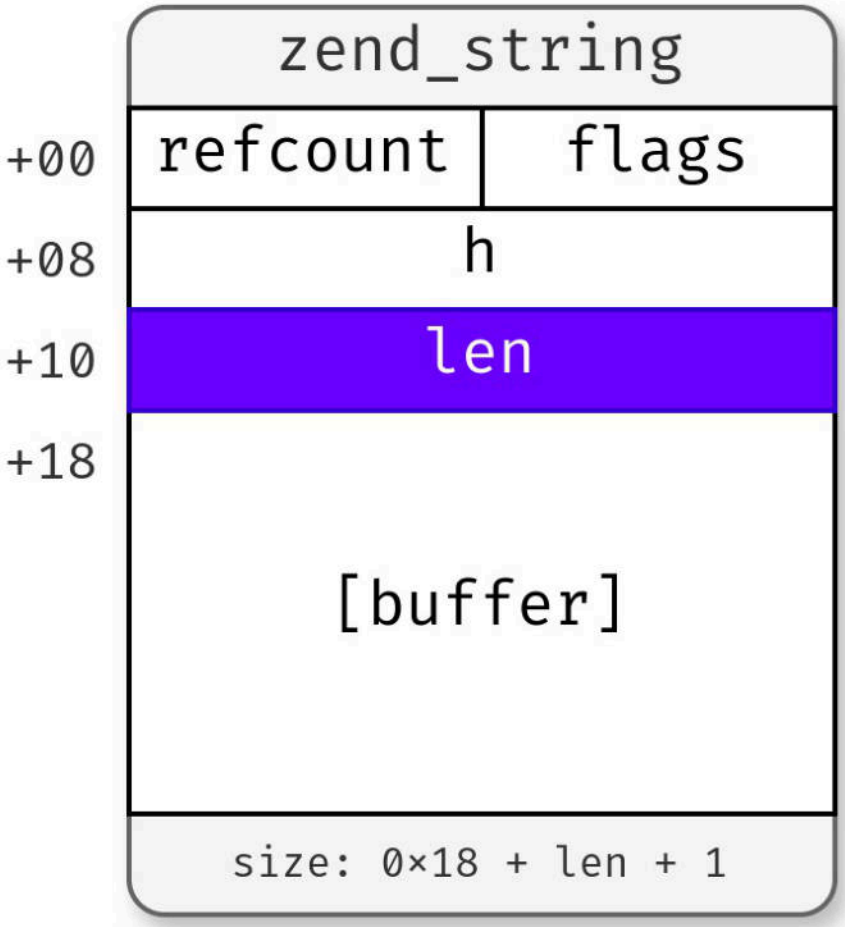
Info leak

- Overwrite → len of a zend_string
- *Before* it is displayed

Bad bug

- 1 to 3 bytes overflow…
- Can only overwrite **refcount**
- Need to **alter free list**

Exploit strategy: getting a leak



Steps

- Corrupt free list
  - Two chunks overlap
- Allocate on top of string
  - Overwrite →len
- String gets displayed

Exploit strategy: reliance on the code

We rely on the code

After we send HTTP parameters, we **cannot** interact with the script anymore

Need to use **« code gadgets »** to perform actions

Exploit strategy: reliance on the code

We rely on the code

```php
<?php

$text = iconv('UTF-8', $_POST['charset'], $_POST['text']);

# end of script

```


Exploit strategy: reliance on the code

We rely on the code

```php
<?php

$text = iconv('UTF-8', $_POST['charset'], $_POST['text']);

$lines = explode("\n", $text); # ← a way to use the bug

```


Real-world exploit

Lexfo

Attacking Roundcube

« Free and open source webmail software for the masses, written in PHP. »

Attacking Roundcube: codewise

When creating an email...

program/include/rcmail_sendmail.php

```php
$charset = rcube_utils::get_input_string('_charset', INPUT_POST);

$mailto  = $this->email_input_format(
    rcube_utils::get_input_string('_to', INPUT_POST, $charset)
);
$mailcc  = $this->email_input_format(
    rcube_utils::get_input_string('_cc', INPUT_POST, $charset)
);
$mailbcc = $this->email_input_format(
    rcube_utils::get_input_string('_bcc', INPUT_POST, $charset)
);

if (!empty($this->invalid_email)) {
    display_error('emailformaterror', 'error', [
        'email' => $this->invalid_email
    ]);
}
```


Attacking Roundcube: memory leak strategy

Simple strategy

1. Set _to to an invalid email ⇒ invalid_email gets set
2. Use _cc to trigger bug ⇒ free list **altered**
3. Use _bcc to allocate on top of invalid_email, altering its size
4. Error is displayed ⇒ **memory leak**

Attacking Roundcube: memory leak failure

A slight oversight

error message is JSON-encoded

```
<script>
 if (window.parent && parent.rcmail) {
  parent.rcmail.iframe_loaded("1");
  parent.rcmail.display_message(
  "Adresse courriel invalide : ajnaajoaajpaajqaa\u0000
  
  
  
  
  
  
  
  
                                                    \u0000MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM\u0
  000\u0001\u0000\u0000\u0000\u0016\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0003\u
  0000\u0000\u0000\u0000\u0000\u0000a@ttest.net
```

Attacking Roundcube: working memory leak strategy

include/rcmail_output_html.php

```php
function get_js_commands(&$framed = null)
{
    $unlock = isset($_REQUEST['_unlock']) ? preg_replace('/[^a-z0-9]/i', '', $_REQUEST['_unlock']) : 0;

    if ($this->framed) {
        $top_commands[] = ['iframe_loaded', $unlock];
    }
    else if ($unlock) {
        $top_commands[] = ['hide_message', $unlock];
    }

    $commands = array_merge($top_commands, $this->js_commands);

    foreach ($commands as $i => $args) {
        $method = array_shift($args);

        foreach ($args as $i => $arg) {
            $args[$i] = self::json_serialize($arg, $this->devel_mode);
        }

        $method = self::JS_OBJECT_NAME . '.' . $method;

        $out .= sprintf("%s(%s);\n", $method, implode(',', $args));
    }

    $framed = $parent_prefix && $parent_commands == count($commands);

    // make the output more compact if all commands go to parent window
    if ($framed) {
        $out = "if (window.parent && parent." . self::JS_OBJECT_NAME . ") {\n"
            . str_replace($parent_prefix, "\tparent.", $out)
            . "}\n";
    }

    return $out;
}
```

Everything is useful

(de)allocations are everywhere

(de)allocations

Made **implode()** of the **second** iteration of the loop overlap with **$out**

Attacking Roundcube: Leak to RCE

RCE is just around the corner

- Data only attack:
  - overwrite **roles**, **serialized** data, ...
- Binary attack:
  - overwrite **zend_array→pDestructor**

Visible part of the iceberg

- Need proper heap setup
- Variable scope is hell
- Very **specific** to server + application
- Generally a huge **pain** to pull off

Attacking Roundcube: DEMO

DEMO

Conclusion

Lexfo

Conclusion: a bad bug... but with some uses

Targeting PHP

- New attack vector: file read to RCE
- Application specific attacks: Roundcube RCE

Other targets

- Other applications may be vulnerable
- Discovered as time go by?

Conclusion: timeline

Timeline (2024)

- **Last year** Crash found
- **February** Started working on bug
- **March 26th** Bug reported to glibc security team
  - They did an amazing job!
- **April 04th** Bug reported to linux distros
- **April 17th** Bug released as **CVE-2024-2961**

Conclusion: resources

Blogposts and exploits

- ambionics.io/blog
- github.com/ambionics/cnext-exploits
- twitter.com/cfreal_

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What ifs

Lexfo

What if: blind file read

```php
~~echo~~ file_get_contents($_GET['file']);
```

What if there is no output?

- No /proc/self/maps
- No access to binaries

What if: no zlib.inflate

What if there is no zlib.inflate? *

- Attack with a single bucket
- Cannot pad heap
- Cannot setup the free list
- Cannot even USE the altered free list

* Very unlikely

Blind, single bucket exploit: DEMO

DEMO

Blind, single-bucket exploit

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